Year 12 Maths - Coordinate geometry

Similar questions - Points and vectors

Two new questions on this topic, a little more open-ended than the exam questions. They are not past exam questions. Answers are at the end.

1 Points A, B and C have position vectors $\mathbf{a} = \binom{1}{2}$, $\mathbf{b} = \binom{4}{6}$ and $\mathbf{c} = \binom{10}{14}$ respectively.

(a) Show that A, B and C lie on a straight line and find the ratio $AB : BC$. **[3]**

(b) The point D has coordinates $(p, -1)$, and triangle ABD is isosceles with $AB = AD$. Find the two possible values of p. **[4]**

2 The line $y = 2x - 3$ passes through the point $P(3,3)$. The point Q also lies on the line, with $Q \neq P$ and $|\overrightarrow{OQ}| = |\overrightarrow{OP}|$, where O is the origin.

(a) Show that the x-coordinate of Q satisfies $5x^2 - 12x - 9 = 0$, and hence find the coordinates of Q. **[4]**

(b) Find the magnitude of $\overrightarrow{PQ}$ in exact form, and the gradient of its direction. **[3]**

(c) The point R is such that $\overrightarrow{PR}$ is perpendicular to $\overrightarrow{OP}$ and $|\overrightarrow{PR}| = \frac{1}{2}|\overrightarrow{OP}|$. Find the position vectors of the two possible positions of R. **[3]**

Question 1

(a) $\overrightarrow{AB} = \binom{3}{4}$ and $\overrightarrow{BC} = \binom{6}{8} = 2\overrightarrow{AB}$. The vectors are parallel and share the point B, so A, B, C are collinear. $AB : BC = 1 : 2$

(b) $AB = \sqrt{9+16} = 5$. $|\overrightarrow{AD}|^2 = (p-1)^2 + (-3)^2 = 25$, so $(p-1)^2 = 16$ and $p = 5$ or $p = -3$

Question 2

(a) $|OP|^2 = 18$. $Q = (x, 2x-3)$: $x^2 + (2x-3)^2 = 18$, so $5x^2 - 12x - 9 = 0$. $(5x+3)(x-3) = 0$; $x = 3$ is P, so $x = -\frac{3}{5}$ and $Q\left(-\frac{3}{5}, -\frac{21}{5}\right)$

(b) $\overrightarrow{PQ} = \binom{-18/5}{-36/5}$, magnitude $\frac{18}{5}\sqrt{5}$. Gradient 2 (parallel to the line)

(c) $|\overrightarrow{OP}| = 3\sqrt{2}$, so $|\overrightarrow{PR}| = \frac{3\sqrt{2}}{2}$. A perpendicular to $\binom{3}{3}$ is $\binom{-1}{1}$, and so $\overrightarrow{PR} = \pm\frac{3}{2}\binom{-1}{1}$. $\mathbf{r} = \binom{3/2}{9/2}$ or $\binom{9/2}{3/2}$